

ZHALTYR, KazakhstanWMO: **351730**Lat: **51.617N**Lon: **69.800E**Elev: **305**StdP: **97.71**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.8	-30.0	-36.4	0.1	-32.6	-33.2	0.2	-29.8	12.0	-7.7	10.1	-8.8	2.0	50	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.3	17.7	30.4	17.4	28.6	17.0	19.6	28.0	18.9	27.3	18.1	26.0	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.6	22.0	16.1	11.8	21.6	15.2	11.2	21.1	57.4	28.0	54.9	27.5	52.4	26.0	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.1	8.3	7.5	DB	-35.6	35.4	2.9	2.0	-37.7	36.8	-39.4	38.0	-41.0	39.1	-43.0	40.5
(5)				WB	-35.6	20.8	2.8	0.9	-37.7	21.5	-39.3	22.0	-40.9	22.5	-42.9	23.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-17.1	-15.3	-7.3	6.0	13.8	18.8	19.8	18.5	12.5	4.8	-5.6	-13.2
	DBStd	14.71	7.67	6.94	6.61	6.39	4.96	4.30	3.77	4.28	5.06	5.35	7.48	7.33
	HDD10.0	3649	839	708	538	149	20	1	0	1	30	176	468	718
	HDD18.3	5784	1098	941	796	369	153	46	28	52	187	419	718	977
	CDD10.0	1122	0	0	0	31	139	267	303	263	104	16	0	0
	CDD18.3	216	0	0	0	1	14	61	73	56	11	0	0	0
	CDH23.3	2537	0	0	0	12	216	741	782	640	144	3	0	0
	CDH26.7	866	0	0	0	1	51	268	269	230	47	0	0	0
Wind	WSAvg	3.2	3.2	3.5	3.6	3.5	3.4	3.1	2.7	2.7	2.8	3.0	3.3	3.4
Precipitation	PrecAvg	332	20	18	20	20	31	40	56	33	21	24	24	25
	PrecMax	533	36	42	46	45	99	99	152	82	54	60	55	56
	PrecMin	201	2	6	4	3	2	10	9	8	2	4	8	6
	PrecStd	71	10	10	13	10	21	24	31	19	14	13	12	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.5	0.8	9.3	24.9	30.8	34.3	34.8	34.1	31.9	22.8	9.6	0.9
		MCWB	-0.3	-0.2	6.1	13.8	16.4	17.8	18.9	18.1	16.3	12.0	6.2	0.2
	2%	DB	-1.5	-1.1	4.4	21.7	27.9	31.7	31.8	31.5	27.1	19.1	6.8	-0.2
		MCWB	-2.2	-2.0	2.6	12.9	15.1	17.2	18.4	17.6	14.6	10.6	4.8	-0.9
	5%	DB	-3.8	-2.9	2.7	19.0	25.7	29.8	29.6	29.3	24.1	16.1	4.6	-1.7
		MCWB	-4.4	-3.6	1.3	11.3	14.3	17.2	18.1	17.4	13.5	9.1	3.0	-2.4
	10%	DB	-5.6	-5.0	1.2	16.0	23.5	27.7	27.6	27.0	21.1	13.0	2.7	-3.3
		MCWB	-6.2	-5.6	0.2	9.6	13.3	16.4	17.7	16.5	12.2	7.8	1.3	-4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	-0.2	6.3	15.2	18.1	20.0	20.7	20.0	17.1	12.9	7.0	0.4
		MCDB	0.4	0.7	9.2	22.5	27.2	28.5	28.8	28.8	29.1	20.4	8.7	0.8
	2%	WB	-2.0	-2.0	2.9	13.3	16.2	19.0	19.9	19.1	15.4	11.1	5.0	-0.8
		MCDB	-1.4	-1.1	4.2	20.8	24.9	28.2	28.1	27.8	24.9	17.7	6.8	-0.3
	5%	WB	-4.3	-3.6	1.4	11.8	15.2	18.1	19.2	18.2	14.2	9.6	3.0	-2.4
		MCDB	-3.9	-2.9	2.5	18.2	23.5	26.9	27.2	26.8	22.7	15.1	4.4	-1.7
	10%	WB	-6.2	-5.6	0.2	10.1	14.1	17.2	18.5	17.3	12.9	8.0	1.4	-4.0
		MCDB	-5.6	-4.9	1.3	15.2	22.0	25.8	25.8	25.4	19.9	12.8	2.6	-3.3
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.3	8.3	10.7	12.7	12.4	11.5	12.4	11.9	9.4	6.9	7.4
		MCDBR	6.9	7.6	6.7	14.9	15.6	15.0	14.1	15.2	15.6	14.4	7.8	7.5
		MCWBR	6.9	7.5	5.5	8.1	6.8	5.7	5.1	5.7	6.8	8.3	6.1	7.1
	5% WB	MCDBR	6.9	7.7	6.5	14.1	13.9	13.2	12.3	13.4	14.4	12.9	7.3	7.1
		MCWBR	6.9	7.6	5.5	8.2	7.0	5.7	5.2	5.8	6.7	8.0	6.0	6.9
Clear-Sky Solar Irradiance	taub	0.292	0.322	0.374	0.407	0.400	0.406	0.425	0.410	0.358	0.342	0.324	0.292	
	taud	2.059	2.014	1.966	2.141	2.232	2.285	2.230	2.254	2.348	2.300	2.151	2.053	
	Ebn at Noon	678	767	797	826	852	849	824	816	823	753	637	602	
	Edh at Noon	80	112	144	137	131	126	131	121	98	84	74	68	
All-Sky Solar Radiation	RadAvg	1.37	2.51	3.88	4.78	5.94	6.30	5.79	5.10	3.79	2.08	0.97	0.90	
	RadStd	0.10	0.10	0.40	0.43	0.53	0.52	0.45	0.35	0.28	0.27	0.11	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A